

CHARAN RAJ KANDULA

Senior Full-Stack Software Engineer

Jersey City, NJ | kandulacharanraj@gmail.com | +1 (551) 256-0939
linkedin.com/in/kandula-charan-raj | github.com/rajkandula | charan-raj-kandula.com

SUMMARY

Full-stack engineer with 6+ years designing and shipping production AI SaaS platforms end to end: React/Next.js and TypeScript frontends, Node.js/Python microservices, and LLM agent infrastructure on GCP and Cloudflare. Owns systems from architecture through deployment, including multi-tenant auth, payments, and serverless, event-driven backends. Focused on scalable system design, distributed systems, and secure AI-agent tooling (MCP).

TECHNICAL SKILLS

Languages & Frameworks: JavaScript, TypeScript, Python, React, Next.js, Node.js, Electron, REST APIs

Cloud & Data: Google Cloud Platform, Cloudflare (Workers, D1, R2, KV, Vectorize), MongoDB, PostgreSQL, Docker, Serverless, CI/CD

AI & Systems: LLM integration, multi-agent systems, MCP, vector search, system design, microservices, event-driven architecture (Kafka, webhooks), OAuth2, RBAC, Stripe payments

EXPERIENCE

Full Stack Engineer · Harakumo · New York, NY Nov 2024 – Present

- Own end-to-end delivery of full-stack features in Next.js/TypeScript on serverless Cloudflare Workers, spanning D1, R2, KV, Vectorize, and Stream, from design to production.
- Designed and built the platform's multi-tenant authentication and authorization layer: organization-level RBAC, API keys, and email OTP verification.
- Implemented Stripe billing end to end: subscriptions, Connect marketplace payouts, and webhook processing for payment lifecycle events.
- Shipped the public developer platform (SDK, CLI, REST API, and documentation) for external partner integrations.
- Built core AI platform features running in production: an AI gateway, agents, and vector/memory stores.

Senior Software Engineer · Emberleaf Books · New York City Metro Jun 2023 – Oct 2024

- Architected and shipped production AI SaaS features across the full stack (Next.js, React, Node.js, Python), owning system design from UI to backend services.
- Designed the multi-agent LLM orchestration pipelines behind the product's AI-assisted content generation in production.
- Built scalable backend microservices with MongoDB on GCP, using Cloudflare for global caching and edge delivery.
- Led cross-platform expansion from web to desktop (Electron, macOS/Windows), reusing shared REST APIs and backend schemas for consistent behavior.

Software Engineer Intern · BibleWrite · New Jersey Jan 2023 – May 2023

- Built Python and Node.js middleware integrating LLM APIs into production features, with React components streaming model responses in real time.
- Implemented request caching and payload optimization to manage API rate limits and reduce token usage.

Graduate Research Assistant (AI/ML) · Stevens Institute of Technology · Hoboken, NJ Jan 2022 – May 2022

- Built Python data-processing pipelines and benchmarked classical ML vs. transformer models on standard NLP benchmarks, measuring accuracy and latency trade-offs.

Software Engineer (Intern to Full-Time) · Chanova Labs · Hyderabad, India Jan 2019 – Dec 2021

- Shipped full-stack features across multiple client projects using React, Node.js, and Python REST APIs with PostgreSQL, including payment integrations, deployed to GCP via CI/CD.
- Converted from intern to full-time; built consumer-facing Android interfaces from Figma wireframes and wrote unit tests.

PROJECTS

Warehouse Sentinel — Autonomous Data-Quality Agents · github.com/rajkandula/warehouse-sentinel Python, Gemini, ADK, GCP

- Built a fleet of background agents that profile every table in a data warehouse, infer data-quality rules with Gemini, validate them, and return a ranked remediation plan (FastAPI on Cloud Run, Firestore checkpoints, Pub/Sub fan-out, BigQuery/DuckDB sources).
- Kept raw rows out of the model by design: each table reduces to a ~2 KB fingerprint via one pushdown aggregate query, with validation as pushdown COUNTIF, so token cost and memory stay flat in table size and customer data never enters a prompt.
- Engineered the guardrails: hard per-job token ceiling, atomic idempotent task claiming so at-least-once redelivery cannot double-charge, per-query byte caps, and checkpoint/resume; verified by a 32-test suite against a planted ground-truth dataset of 7 defect classes.

EDUCATION

M.S. in Computer Science · Stevens Institute of Technology, Hoboken, NJ Jan 2022 – May 2023

B.Tech in Computer Science · Mahatma Gandhi Institute of Technology, Hyderabad, India Jun 2016 – Oct 2020